Signal Conduction

Note: In OneNote, set this worksheet as background. Use "Export" in the simulation and insert the image in the marked space.

At the axon hillock, the signals previously received through dendrites and the cell body are brought together. If a threshold is reached, an electrical impulse is generated and passed into the axon. The axon is a long extension through which information can be transported over longer distances. Many axons are surrounded by a sheath cell. The sheath cell acts as an electrical insulator. The nodes lie between its sections. At these nodes, the electrical signal is rebuilt each time, so that the excitation appears to jump from node to node. This form of conduction is called saltatory conduction. An unmyelinated axon does not have such an insulating covering. Therefore, the excitation must instead be passed on again at every neighbouring membrane section. This continuous conduction is slower.

Conduction along the axon ensures that information quickly reaches the terminal buttons from the cell body. There, the signal can be transmitted to the next cell.

Task 1: Comparing saltatory and continuous conduction

Compare conduction in the upper and lower axon. Describe how you can tell that excitation is conducted saltatorily in the upper axon and continuously in the lower axon.

Task 2: Explaining conduction

Using technical terms, explain why the myelin sheath speeds up conduction.

Task 3: Model experiment on conduction

Build two models of signal conduction in neurons: one for continuous conduction and one for saltatory conduction. Use the dominoes and straws provided and photograph your models.

Image:

Model of continuous conduction

Image:

Model of saltatory conduction